

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA1601

COURSE NAME: Data Warehousing & Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision

support systems. BL1, BL2

Assessment Tool Weightage: 35%

QUESTIONS: 5 Total Marks: 50

Annexure A

CONCEPT MAPPING Duration :

Code of Conduct:

I Tharun Sai Gangadhar P (192521486) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Tharun Sai . P

Annexure A

Concept Map 1: DSS and Business Intelligence Integration

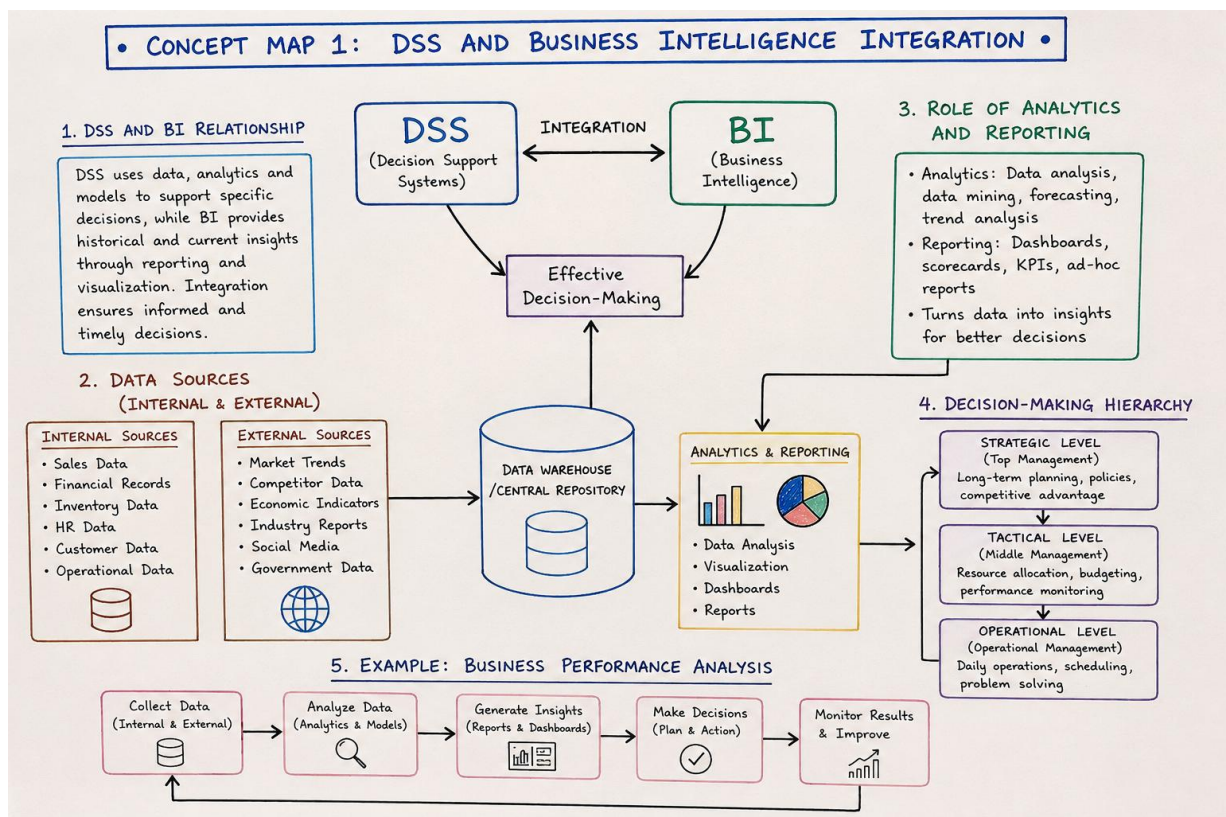
Task Description:

Construct a concept map explaining how Decision Support Systems integrate with Business Intelligence for effective decision-making.

What your map must show:

1. DSS and BI relationship
2. Data sources (internal & external)
3. Role of analytics and reporting
4. Decision-making hierarchy
5. Example: Business performance analysis

ANSWER :



Concept Map 2: Data Processing Ecosystem (OLTP → OLAP → BI)

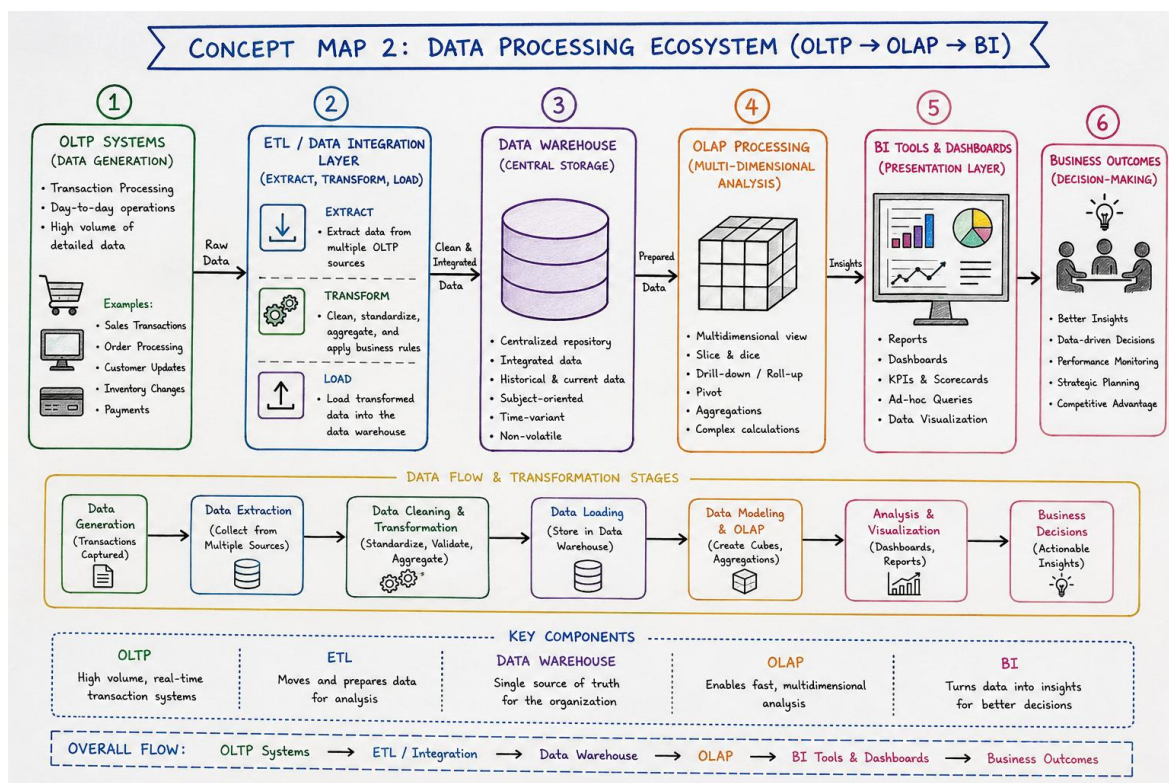
Task Description:

Create a concept map illustrating the complete data processing ecosystem from transaction systems to analytics.

What your map must show:

1. OLTP systems (data generation)
2. ETL/data integration layer
3. Data warehouse storage
4. OLAP processing
5. BI tools and dashboards
6. Data flow and transformation stages

ANSWER :



Concept Map 3: Data Warehouse Components and Data Flow

Task Description:

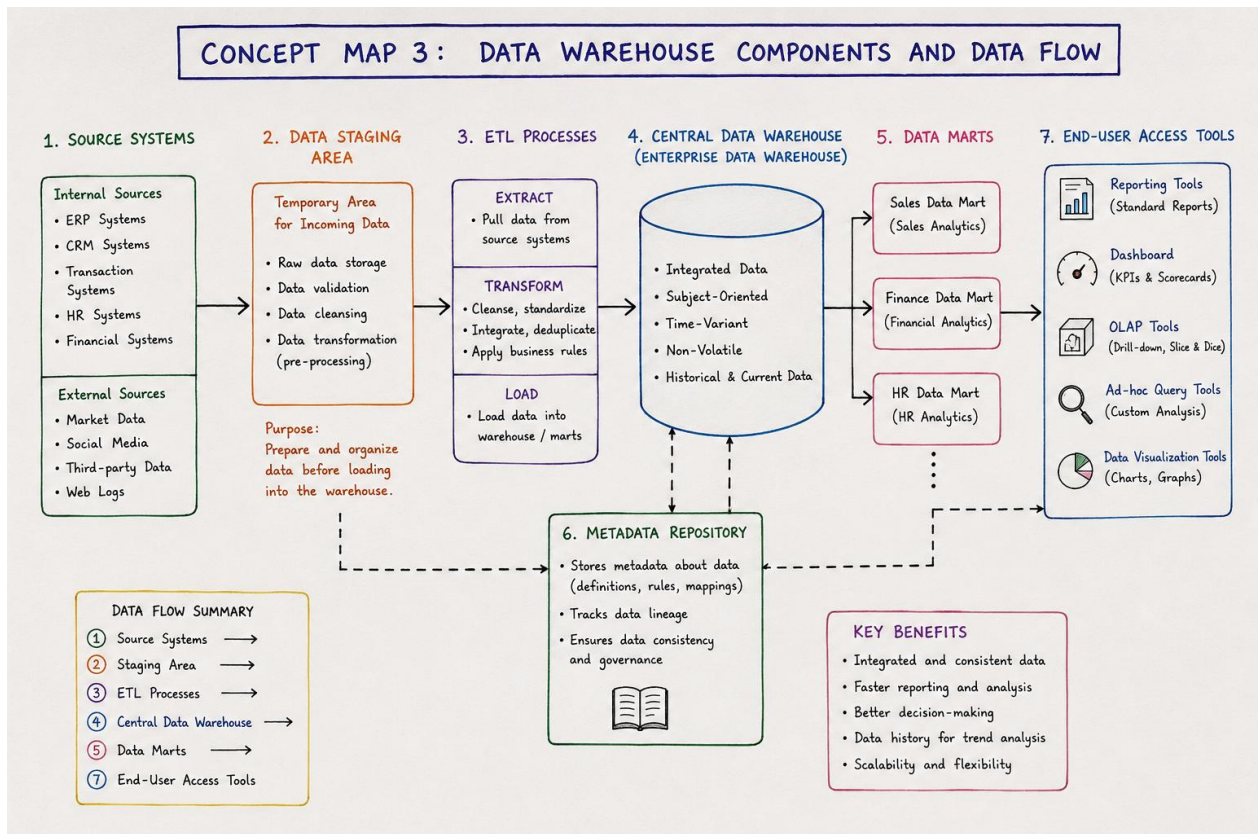
Design a concept map highlighting the internal components of a data warehouse and how data flows

through them.

What your map must show:

1. Source systems
2. Data staging area
3. ETL processes
4. Central warehouse
5. Data marts
6. Metadata repository
7. End-user access tools

ANSWER :



Concept Map 4: Dimensional Modelling and Schema Design Strategy

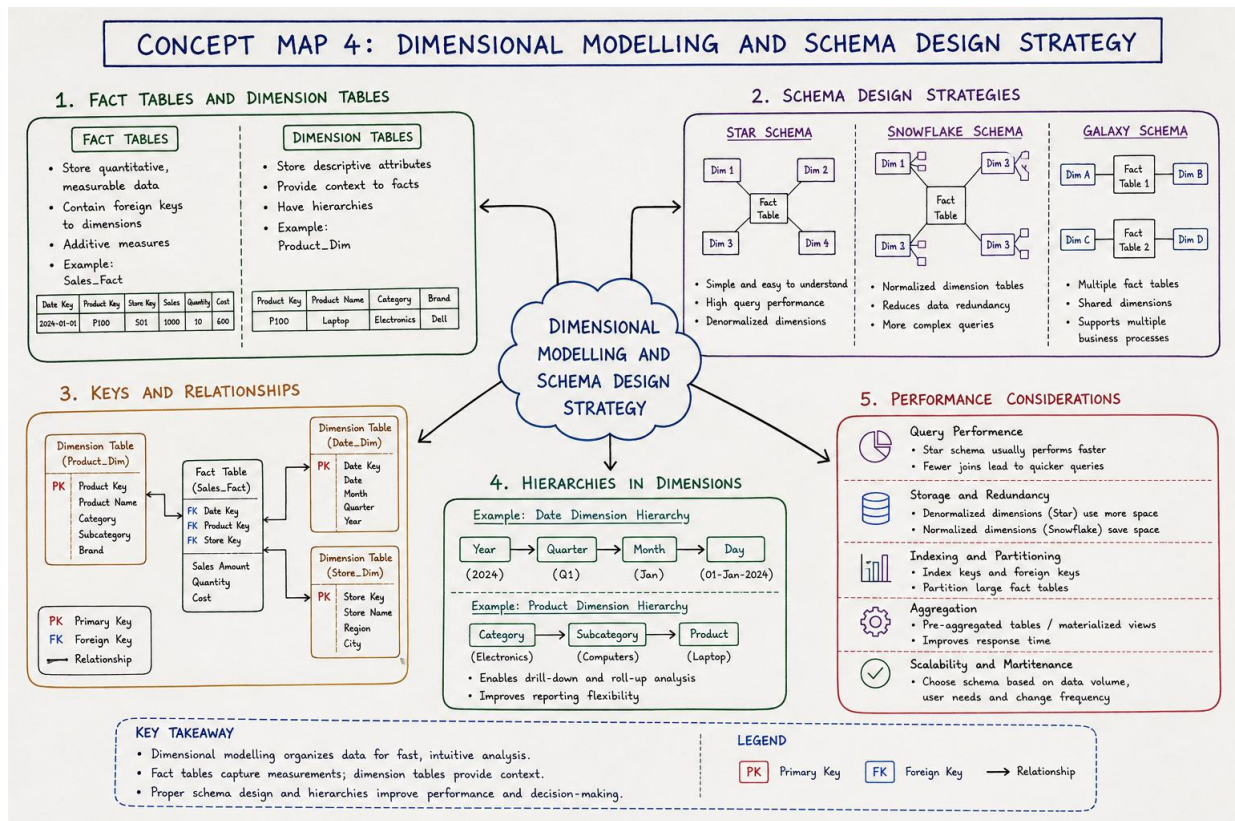
Task Description:

Create a concept map focusing on schema design strategies in data warehousing.

What your map must show:

1. Fact tables and dimension tables
2. Star, snowflake, and galaxy schemas
3. Keys and relationships
4. Hierarchies in dimensions
5. Performance considerations

ANSWER :



Concept Map 5: KNIME Analytics Workflow

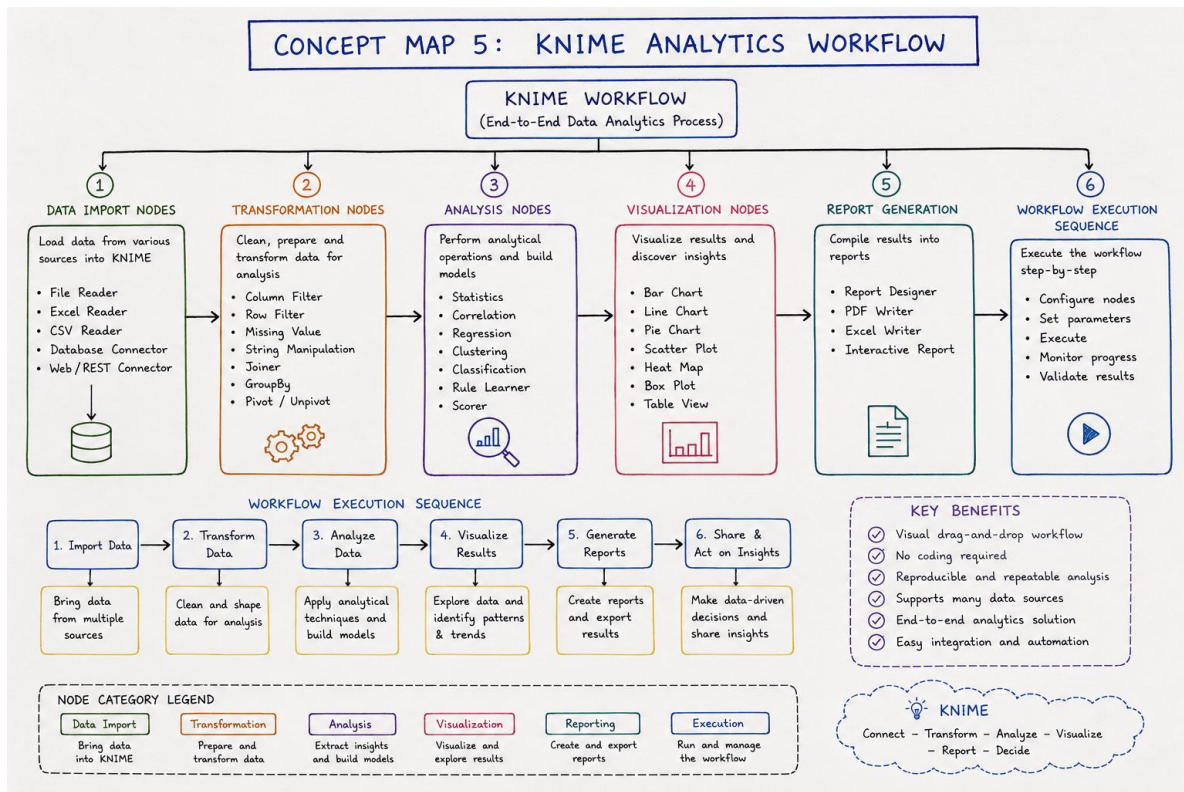
Task Description:

Create a concept map that explains the workflow of data analytics using KNIME.

What your map must show:

1. Data import nodes
2. Transformation nodes
3. Analysis nodes
4. Visualization nodes
5. Report generation
6. Workflow execution sequence

ANSWER :



Concept Map 6: Advanced OLAP Operations Map

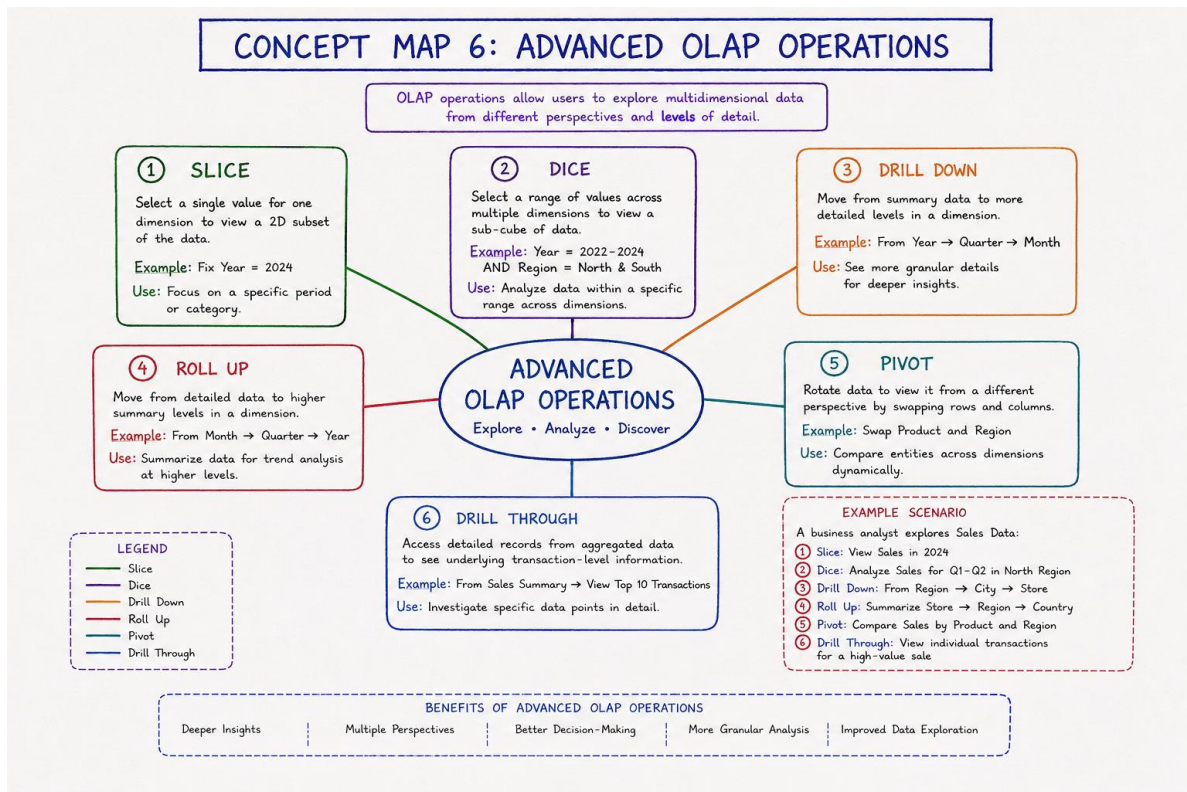
Task Description:

Develop a concept map explaining advanced OLAP operations and their relationships.

What your map must show:

1. Slice
2. Dice
3. Drill Down
4. Roll Up
5. Pivot
6. Drill Through

ANSWER :



Concept Map 7: Data Quality and Preprocessing Map

Task Description:

Create a concept map explaining the importance of data quality in BI systems.

What your map must show:

1. Missing Data Handling
2. Noise Reduction
3. Data Transformation
4. Data Integration
5. Impact on Decision Making

ANSWER :

CONCEPT MAP 7: DATA QUALITY AND PREPROCESSING MAP

High quality data is the foundation of reliable Business Intelligence. It ensures accurate analysis, trustworthy reports and better decisions.

DATA QUALITY & PREPROCESSING

Raw data from multiple sources often contains issues. Preprocessing improves data quality and prepares it for analysis and decision-making.

1 MISSING DATA HANDLING

- Identify missing values
- Analyze pattern of missingness
- Handle using appropriate techniques

Techniques:

- Deletion (listwise / pairwise)
- Imputation (mean, median, mode)
- Prediction-based imputation

A	B	C	A	B	C
1	2	-	1	2	5
4	-	6	4	6	6
7	8	9	7	8	9

2 NOISE REDUCTION

- Detect and remove noise and outliers
- Smoothen data for consistency
- Improve reliability of data

Techniques:

- Outlier detection
- Binning
- Smoothing (moving avg.)
- Filtering

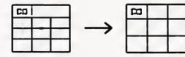


3 DATA TRANSFORMATION

- Convert data into a suitable format for analysis
- Normalize and standardize data
- Create new attributes
- Aggregate data when required

Techniques:

- Normalization (Min-Max, Z-score)
- Aggregation (sum, avg., count)
- Encoding (categorical to numeric)
- Data type conversion

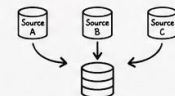


4 DATA INTEGRATION

- Combine data from multiple sources
- Resolve conflicts and inconsistencies
- Ensure unified and consistent view

Methods:

- Schema integration
- Entity resolution
- Data matching
- Redundancy removal



5 IMPACT ON DECISION MAKING

- High quality data leads to:
 - ✓ Accurate insights
 - ✓ Reliable reports
 - ✓ Better forecasting
 - ✓ Efficient operations
 - ✓ Increased trust in BI system



HOW DATA QUALITY IMPROVES BI OUTCOMES



COMMON DATA QUALITY ISSUES

- Incomplete data
- Inaccurate data
- Inconsistent data
- Duplicate data
- Outdated data
- Irrelevant data

Good data in → Good information out → Good decisions → Better business performance.